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Theoretical Investigation of Dosimeter Accuracy for Linear Energy Transfer Measurements in Proton Therapy: A Comparative Study of Stopping Power Ratios

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Abstract:	Background Accurate measurement of linear energy transfer (LET) is crucial in medical physics, particularly for proton therapy dosimetry. High atomic number ([[EQUATION]]) materials like BaFBr and low [[EQUATION]]materials like Al2O3 and water are commonly used in dosimeters. Purpose To evaluate the feasibility and accuracy of using various dosimetry materials (water, air, Al2O3, aluminum (Al), BaFBr, and oxygen) in measuring LET by comparing their stopping power ratios using the Bethe-Bloch theory and semi-empirical models. Methods Stopping power ratios were calculated using the PSTAR database for proton energies ranging from 0.01 MeV to 10,000 MeV. The Bethe-Bloch theory with density and shell corrections was used for high-energy protons, while a semi-empirical model was applied for low-energy protons. Experimental validation involved comparing the theoretical stopping power data with SRIM and PSTAR data for materials including water, aluminum, air, Al2O3, BaFBr, and oxygen. Results The stopping power ratio of water to air at 10 MeV was found to be approximately 1.05. The water-to-Al2O3 ratio at 1 MeV was approximately 1.12, and the BaFBr-to-water ratio was around 0.95. These findings suggest a high degree of accuracy in LET measurement using these materials. The comparison of theoretical models with experimental data showed strong agreement, particularly for high-energy protons where the Bethe-Bloch theory was applied. Conclusions This study confirms the feasibility of using high [[EQUATION]] materials like BaFBr and low [[EQUATION]]materials like Al2O3 and water for accurate LET measurement in proton therapy. The results validate the use of the Bethe-Bloch theory and semi-empirical models in dosimetric applications, enhancing the precision of LET measurements and contributing to improved radiation therapy outcomes.

Highlights

- Evaluation of dosimetry materials for accurate linear energy transfer (LET) measurement.
- Utilization of Bethe-Bloch theory with density and shell corrections for high-energy protons.
- Application of a semi-empirical model for low-energy proton LET calculations.
- Validation of theoretical models with experimental data from SRIM and PSTAR.
- Findings support the use of high Z_eff materials like BaFBr and low Z_eff materials like Al2O3 and water in proton therapy dosimetry.

Theoretical Investigation of Dosimeter Accuracy for Linear Energy Transfer Measurements in Proton Therapy: A Comparative Study of Stopping Power Ratios

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Abstract

Background: Accurate measurement of linear energy transfer (LET) is crucial in medical physics, particularly for proton therapy dosimetry. High atomic-number (Z_{eff}) materials such as BaFBr and low-Z_{eff} materials such as Al2O3 and water are commonly used in dosimeters.

Purpose: To evaluate the feasibility and accuracy of the use of various dosimetry materials (water, air, Al2O3, aluminum (Al), BaFBr, and oxygen) for measuring LET by comparing their stopping power ratios via the Bethe-Bloch theory and semiempirical models.

Methods: Stopping power ratios were calculated via the PSTAR database for proton energies ranging from 0.01 MeV to 10,000 MeV. The Bethe-Bloch theory with density and shell corrections was used for high-energy protons, whereas a semiempirical model was applied for low-energy protons. Experimental validation involved comparing the theoretical stopping power data with the SRIM and PSTAR data for materials such as water, aluminum, air, Al2O3, BaFBr, and oxygen.

Results: The stopping power ratio of water to air at 10 MeV was found to be approximately 1.05. The water-to-Al2O3 ratio at 1 MeV was approximately 1.12, and the BaFBr-to-water ratio was approximately 0.95. These findings suggest a high degree of accuracy in LET measurement using these materials. A comparison of the theoretical models with the experimental data revealed strong agreement, particularly for the high-energy protons where the Bethe-Bloch theory was applied.

Conclusions: This study confirms the feasibility of using high-Z_{eff} materials such as BaFBr and low-Z_{eff} materials such as Al2O3 and water for accurate LET measurements in proton therapy. The results validate the use of the Bethe-Bloch theory and semiempirical models in dosimetric applications, enhancing the precision of LET measurements and contributing to improved radiation therapy outcomes.

Keywords: Linear Energy Transfer, Dosimetry, Bethe-Bloch Theory, Proton Therapy, Stopping Power, PSTAR

1. INTRODUCTION

In the field of medical physics, precise dosimetry is essential for the effective application of proton therapy. Dosimeters that measure LET accurately are critical for ensuring that the correct dosage is administered to patients. This study focuses on evaluating the accuracy of dosimeters made from materials with different Z_{effs} and uses theoretical and experimental data to assess their performance.

In recent years, interest in both theoretical and experimental studies of the stopping power and particle range of various materials has increased. Charged particles lose energy through constant collisions when they contact matter. The stopping power is determined by the particle's type and energy, as well as the characteristics of the substance it travels through. The material's stopping power is related to the ionization density along the path since creating an ion pair requires a specific amount of energy. The stopping power refers to the material's characteristics, whereas the energy loss per unit path length is related to the particle's interactions.

An accurate understanding of the stopping powers of different materials is crucial for radiation dosimetry calculations, radiation quality evaluations, and modeling of energy deposition patterns. To incorporate fresh experimental data and refinements in analytic methods, stopping power data must be reviewed on a regular basis. In addition, calculating the dosage at which ionizing particles deposit requires stopping powers. The International Atomic Energy Agency's (IAEA) (TRS-398) most recent proton dosimetry protocol [1] establishes a standard for proton dosimetry and addresses dosimetry for ions heavier than protons.

Linear energy transfer (LET) is crucial for assessing biological efficacy and radiation quality and it is tied to both unconstrained and restricted electronic stopping powers. An accurate understanding of energy loss mechanisms and rates in diverse materials is essential for determining the impact of radiation on biological systems and improving therapeutic regimens. For high energies (above 1 MeV for protons and 10 keV for electrons), Bethe's stopping-power theory is recommended for determining the collision stopping power [6]. This theory relies on the mean excitation energy, which is difficult to estimate theoretically for complex materials and requires experimental input.

At energies below 100 keV/nucleon, Bethe theory is less effective, making experimental data the main source, although it is often insufficient or inconsistent. Empirical fitting formulas such as Andersen-Barkas and Ziegler's shell corrections are used to extrapolate data. Experimental stopping power data are available for elemental targets in databases such as PSTAR and SRIM (2008) but not for complex materials.

Bragg's rule helps estimate the stopping power of compounds and mixtures via constituent element data, simplifying computations. BaFBr is a promising detector material for high-LET radiation because of its strong radiation hardness and quick scintillation response, making it suitable for high-precision dosimetry applications (Woody, 1992 [7]).

This study aims to evaluate the accuracy of various dosimetric materials in measuring LET by comparing their stopping power ratios via theoretical models and simulations.

2. Methods

Theoretical Framework

The stopping power calculations for high proton energies are based on the Bethe-Bloch theory, which accounts for the energy loss mechanisms in different materials. To improve the accuracy of these calculations, density and shell corrections are incorporated. For low proton energies, where the Bethe-Bloch theory becomes less accurate, a semiempirical model is employed.

Materials

The studies focused on the common dosimetry materials investigated include the following:

- **Water:** A reference material in dosimetry.
- **Air:** Commonly used in ionization chambers.
- **Al2O3 (Alumina):** Utilized in thermoluminescent dosimeters.
- **BaFBr:** A component of imaging plates in computed radiography.
- **Oxygen:** Relevant owing to its biological importance.

Simulation and Validation

To verify the accuracy of our models, the stopping power ratios for the materials specified were calculated theoretically and compared with SRIM (2008) simulations and experimental PSTAR data. These ratios were compared across different materials to evaluate the dosimeter performance in various radiation environments.

The accuracy of LET measurements was assessed by comparing the theoretical predictions with experimental data.

Experimental Procedure

Stopping power ratios were calculated via the PSTAR database for proton energies from 0.01 MeV to 10,000 MeV. The Bethe-Bloch theory with density and shell corrections was used for high-energy protons, whereas a semiempirical model was employed for low-energy protons. Experimental validation involved comparing theoretical data with the SRIM and PSTAR data for the specified materials.

Fano proposed the following formula:

$$S = \frac{4\pi e^4 Z_2}{m_e v^2} Z_1^2 \left[In \frac{2mv^2}{\langle I \rangle} - In(1-\beta^2) - \beta^2 - \frac{C}{Z_2} - \frac{\delta}{2} \right] \tag{1}$$

The Bethe-Bloch equation (after 20 years) is based on three key assumptions:

1. The ion has been completely stripped (> 1 MeV/nucleon): This assumes that the incident ion is fully ionized, meaning that it has lost all its orbital electrons at high enough energies (over ~1 MeV per nucleon). Treating the ion as a bare point charge simplifies the calculation.
2. The ion travels at a quicker speed than the orbital electrons of the target. The derivation is based on the premise that the incident ion is moving faster than the orbital electrons of the target material. This enables one to consider the target electrons as though they are roughly at rest.
3. The ion weighs much more than the electrons in the target: the ion is regarded by the Bethe-Bloch equation as being significantly more massive than the target material's electrons. This makes it possible to ignore ion recoil during collisions with atomic electrons.

Low-Energy Boundary of the Bethe-Bloch Theory: Neutralization of Particles

The Bethe-Bloch theory's low-energy boundary solves the theory's limitations in regard to particles traveling at slower speeds. Particles are more likely to pick up electrons and become neutralized at these lower energies, which drastically changes their stopping power. To precisely represent the energy loss and interactions of these particles, modified theoretical models, such as Ziegler's shell, the density effect, and Andersen-Barka's and Bloch corrections, must be used. Comprehending this limit is crucial for precise forecasts and use in the field of radiation dosimetry.

Empirical relation for stopping power

Ziegler (1999) [9] established an empirical relation expressed as follows:

$$S = \frac{KZ_2}{\beta^2} Z_1^2 \left\{ \left[f(\beta) - \frac{C}{Z_2}(Shell) - In\langle I \rangle - \frac{\delta}{2}(Density) \right] + Z_1 L_1 (Barkas) + Z_1^2 L_2^2 (Bloch) \right\} \quad (2)$$

The primary stopping contributions are from, $In\langle I \rangle$, and the particle velocity, $f(\beta)$, and are derived originally from Bethe–Bloch, as shown in Eq. (1). There is less pausing since the subsequent phrase, $In\langle I \rangle$, is negative. The three correction factors that are essential at low energy in calculating the stopping power are (C/Z_2), $Z_1 L_1$ and $Z_1^2 L_2^2$.

At all energies, the Bloch correction accounts for less than 1% of the halting. The density correction $\delta/2$ is the sole significant correction term for very high energies; for energies below 1 GeV/u, it contributes less than 1%.

The shell correction term $\frac{C}{Z_2}(\boldsymbol{Shell})$

For proton energies less than 1 MeV, the shell correction term is very important since it can significantly affect how accurately stopping power calculations are performed in this energy range. Ignoring this correction might result in substantial departures from experimental data—particularly in the case of complicated electronic structures or materials with high ionization potentials. One of the fundamental presumptions of the original Bethe formula, the collapse of the dipole approximation, is considered by this adjustment. Concerns about shell corrections have existed for some time, and early investigators understood that orbital-by-orbital adjustments were necessary. Usually, it is estimated by carefully examining how the particle interacts with every electronic orbit in different elements. A correction of up to 10% can be attributed to stopping powers. Since the discrete energy levels of atomic electron shells greatly affect the energy loss, shell corrections are especially important for low energy charged particles. In 2009, Sabin and Oddershede [12] selected an ansatz for the shell correction form:

$$\frac{C(v)}{Z_2} = C_1 v e^{-c_2 v} \quad (3)$$

C1 and C2 were obtained by fitting the computed shell correction values for H, C, N, O and other elements to Eq. 3 to derive the shell corrections [13]. Table 1 provides that information [12].

Table 1: Shell Correction Parameters for Use in Equation 3 [12]

Atom	C1	C2
H	1.50	1.00
C	0.14	0.20
N	0.15	0.21
O	0.26	0.33
He	1.02	0.67
Be	0.21	0.25
F	0.32	0.28
Li	0.29	0.29
Ne	0.30	0.28
Al	0.18	0.31

The mean ionization potential, $\boldsymbol{In\langle I\rangle}$

A key component of the Bethe-Bloch formula is the mean ionization potential, or $\langle \boldsymbol{I} \rangle$. This is because, particularly at lower particle energies where the logarithmic term becomes more important, it directly affects the logarithmic term that controls the energy loss rate. Precise energy loss calculations depend on the correct determination of energy loss. Precise values of $\langle \boldsymbol{I} \rangle$, which are determined through experimental data, are critical for dosimetry and for anticipating the stopping power of various materials, which is vital in radiation therapy.

The density effect correction term $\frac{\delta}{2}(\boldsymbol{Density})$

Raymond Sternheimer et al. [8] developed a comprehensive theoretical framework in 1984 to explain the density effect correction. This adjustment ensures that the Bethe-Bloch equation accurately represents the polarization effects occurring at high particle velocities by modifying the logarithmic term. At high velocities, the electromagnetic field of a traveling charged particle polarizes the atoms in the medium, affecting the stopping power. This correction is an essential term in the Bethe-Bloch formula, which reduces the energy loss rate of high-energy charged particles in dense materials by considering the medium's polarization. To achieve accurate stopping power calculations, especially in dense materials and at high velocities, the Bethe-Bloch formula must be incorporated $\delta/2$. This correction is crucial for applications requiring precise modeling of particle interactions with matter, such as in the field of radiation dosimetry.

We applied the following equation ref [8]:

```
def density_corrections(kinetic_energy, C, X0, X1, a, m):

    gamma = (1 - (proton_beta(kinetic_energy)) ** 2) ** (-0.5)

    X = np.log10(proton_beta(kinetic_energy) * gamma)

    if X < X0:

        return 0

    if X > X0 and X < X1:

        return 4.6052 * X + a * (X1 - X) ** m + C

    if X > X1:

        return 4.6052 * X + C
```

For precise computations, the parameters C, X0, X1, a, and m must be supplied, and these are material-dependent parameters.

Empirical Barka's correction term Z_1L_1

Barka's correction in the Bethe-Bloch formula explains the variation in the stopping power (energy loss rate), and Z_1L_1 is a crucial term. In applications such as radiation therapy and in the field of dosimetry, where precise dose delivery is necessary for efficient treatment, the Barka's correction is vital for attaining high precision in stopping power estimates. By ensuring that the models utilized in particle physics and dosimetry more closely reflect the real behavior of particles in a medium, the accuracy of the measurements and simulations can be increased. For target materials with greater atomic numbers and at lower particle velocities, Barka's adjustment becomes more relevant.

We applied the following Barka's correction developed by Ziegler, 1999 [16].

```
def barkas_correction(T, Z):

    T_eV = T * 1000

    L_low = 0.001 * T_eV

    L_high = (1.5/(T_eV**0.4)) + (45000/Z)/(T_eV**1.6)

    barkas = L_low * L_high/(L_low + L_high)

    return barkas
```

For precise stopping power estimates, it is necessary to determine the Barka's correction, which can be easily computed via the above function for a given kinetic energy and atomic number. Experimental data and theoretical models are the sources of the formulas utilized in the Barka's correction function. The objective of these formulas is to depict how Barka's correction depends on the kinetic energy and charge of the particle, in addition to the characteristics of the target material (charge Z). The difference in the stopping power between positive and negative particles of the same velocity can be considered via the Barka's correction term $\boldsymbol{Z_1 L_1}$, which is computed via this function in conjunction with other elements of the Bethe-Bloch formula. With lower particle velocities and larger atomic number target materials, the significance of this adjustment increases.

Bloch's correction $\boldsymbol{Z_1^2 L_2^2(Bloch)}$

The Bethe-Bloch formula, which is derived via perturbation theory, is proportional to the square of the incident particle's charge (z^2), which is explained by the Bloch correction. At lower particle energies, when the particle velocity is not extremely relativistic, the Bloch correction becomes relevant.

We applied the following Bloch's correction from Ziegler, 1999 [16], which is a straightforward equation that was proposed by Bichsel from the practical perspective of computing accurate stopping powers. In particular, the work of Wyckoff, 1993 [9], as cited in @ICRU-49], and previous discoveries by Bichsel (1990) [10] are as follows:

```
def bloch_correction(kinetic_energy):

be = proton_beta(kinetic_energy)

y = 1/(137 * be)
```

```
bloch = -y**2 * (1.202 - y**2 * (1.042 - 0.855 * y**2 + 0.343 * y**4))

return bloch
```

To precisely determine the stopping power of charged particles, especially at low energies where the Bethe-Bloch formula is insufficient, the Bloch correction factor, as shown above, is usually utilized in combination with other factors in the formula.

3. RESULTS

3.1 Analytical shell corrections

When determining the stopping power of charged particles in matter, shell correction is crucial, especially when the energy of the particles is low. It has been suggested that shell adjustments be included in stopping power estimations through several analytical representations. In 2009, by Sabin and Oddershede [12], one such analytical representation was provided. To express the shell corrections to the stopping power in an analytical expression, they suggest an ansatz, which is a hypothesis or an educated estimate. Their ansatz performs well in capturing the data and offers a manipulable mathematical formula for entire atom-shell corrections, which is helpful when calculating the stopping power. We used the information from Ref. [12], to calculate the coefficients of the atoms to test the mechanism.

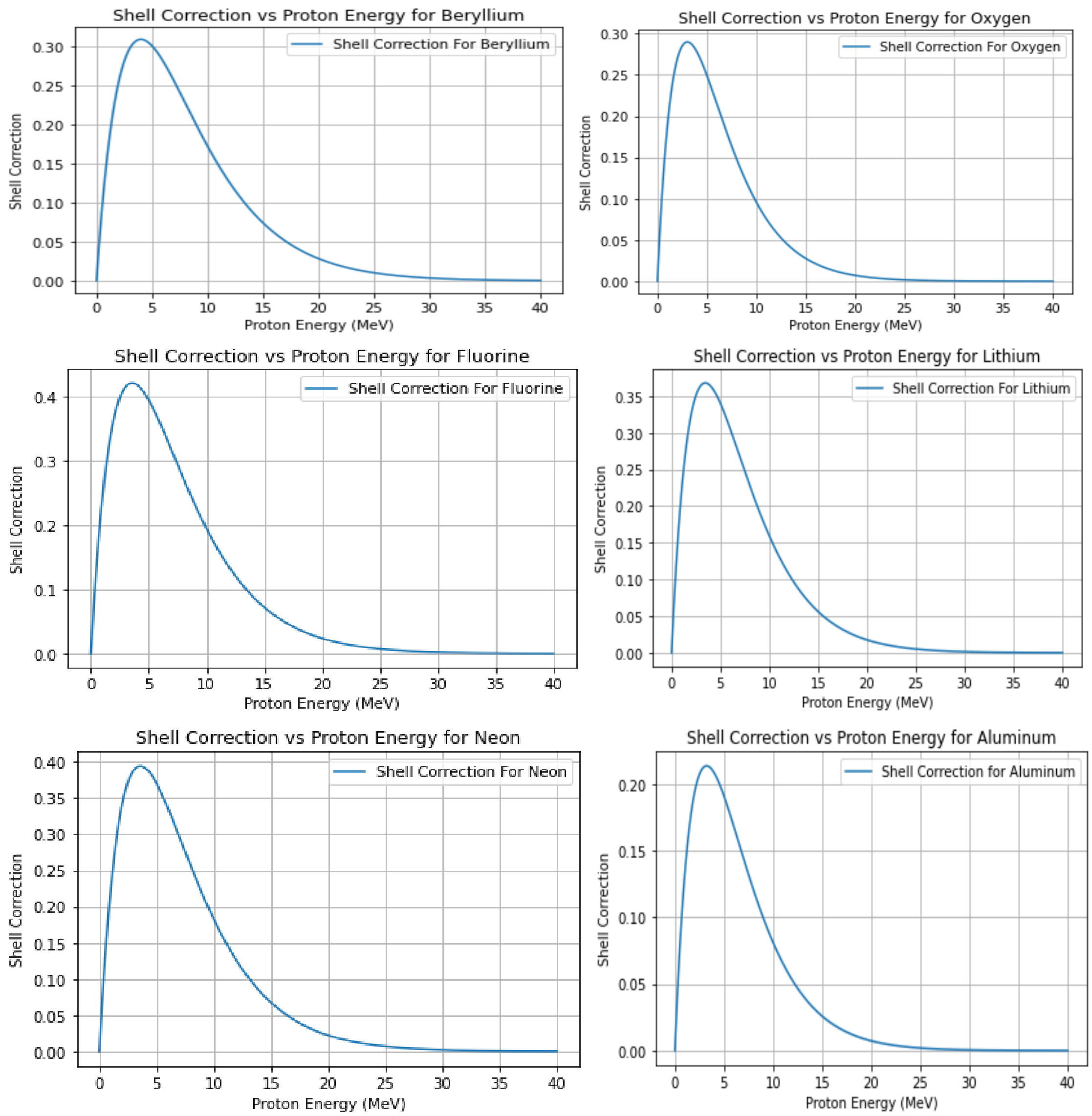

Figure 1: Shell corrections for beryllium, fluorine, lithium, neon, oxygen, and aluminum are shown in this plot as a function of proton energy, which was calculated from Eq. (3) and Ref. [11]. Through a comparison of the shell corrections derived from Ref. [11] and Eq. (3), any disparities or convergences in the data can be identified, offering valuable information on the precision and dependability of the theoretical model embodied by Equation (3). Typically, an orbital-by-orbital method is employed to compute shell adjustments. The projectile velocities vary where the shell corrections peak due to fluctuations in orbital electron velocities, which are considered via this method.

3.2 Analytical Proton Stopping Power Compared with PSTAR Data and SRIM-2008 Data

The stopping power is predicted analytically via mathematical models and equations. The Bethe-Bloch formula, as shown in equation (2), which explains the stopping power of charged particles as they move through matter, is one such concept. Analytical models or Monte Carlo simulations can be used to calculate the proton stopping power, which measures the energy loss of protons per unit route length as they pass through a material. The proton velocity and target material characteristics, such as density and ionization state, affect the stopping power. Despite several assumptions and approximations, the analytical models can yield credible stopping power estimates compared with experimental data (PSTAR) [15] and Monte Carlo simulations (SRIM-2008) [17]. We have calculated the stopping powers for various elements and compounds by using the relation provided in equation (2) and accurately applying other empirical equations. The determination of a proton's energy loss per unit distance through a substance is known as the analytical proton stopping power. In fields such as materials science, nuclear physics, and medical physics, this computation is essential. For a given material, the stopping power can be calculated as follows: find the atomic mass (A), atomic number (Z), and mean excitation energy (I) of the material. To find the overall stopping power, apply the Bethe-Bloch equation (2) is applied, the required adjustments (density and shell effects) are made, and the nuclear and electronic stopping powers are summed.

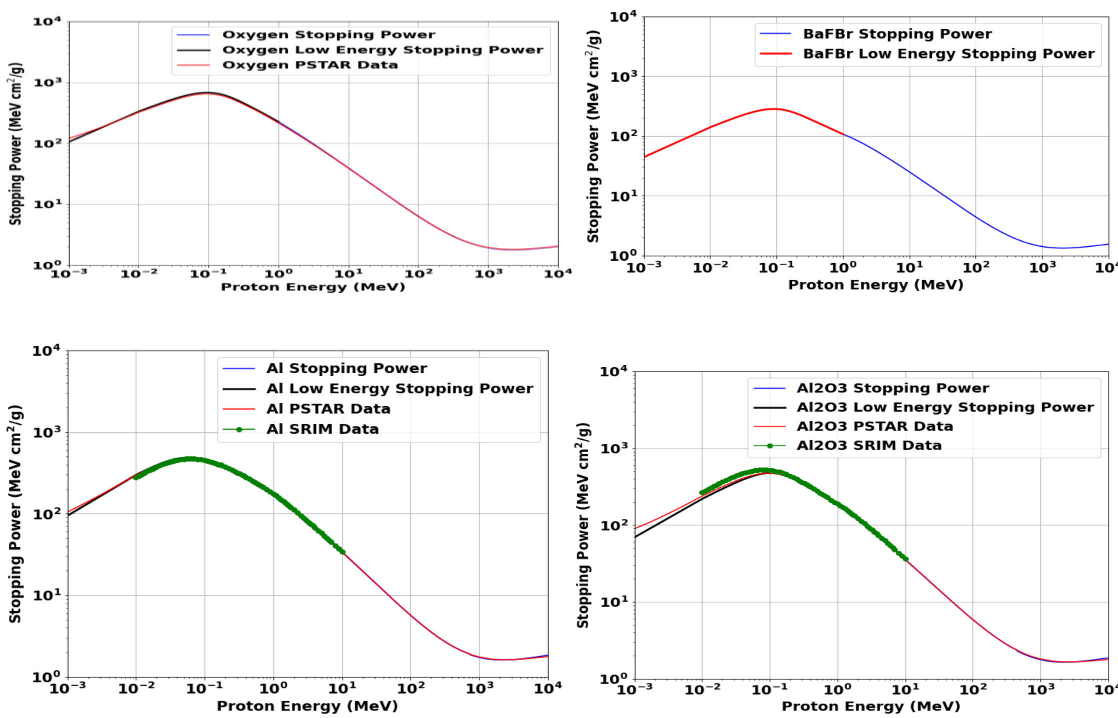

Figure 2: Comparison of theoretical model predictions with SRIM and PSTAR data for selected materials. Our theoretical models strongly agreed with both the SRIM simulations and the PSTAR experimental data, particularly for high-energy protons where the Bethe-Bloch theory was applied. The proton stopping power calculations for different elements and compounds are shown in this figure, which contrasts them with the relevant data from the Pstar database and SRIM-2008. The calculations were performed analytically. This comparison involves the elements oxygen (O), aluminum (Al), and a compound Al_2O_3. The stopping power is plotted against the proton energy, which ranges from 0.1 MeV to 10000 MeV and is represented in $MeVcm^2/g$. Using the Bethe-Bloch formula, the analytical computations were carried out to ensure correctness at different proton velocities by integrating the required corrections, such as those of Barka's, Ziegler's shell, density effects, and Bloch. The Pstar data [15] and SRIM-2008 data [17], shown in the figure, act as a reference points for confirming the analytical findings. This comparison illustrates how well the theoretical model and actual data match, showing how useful the analytical method is for estimating the proton-stopping power in a variety of materials. For every element and compound, the overall trend and size of the stopping power particular energy bands where the Pstar and SRIM-2008 data and the analytical estimates strongly agree with how the atomic number, mean excitation energy and density affect the stopping power of a material. **Observations:** Robust modeling is observed in the smooth transition between the low-energy and high-energy models, and the theoretical stopping power curve (blue line) closely aligns with the experimental data (red and green) quite well.

3.3 Evaluation of the stopping power ratio

The quantitative analysis indicates that the water to air stopping power ratio at 10 Mev is about 1.05 and this shows how accurate LET measurements are. The ratio of water-to-Al2O3 was determined to be about 1.12 at 1 MeV, whereas BaFBr-to-water ratio was about 0.95. The agreement between the measured and theoretical values especially for high-energy protons affirms the trustworthiness of Bethe-Bloch theory. There are several important issues and steps in radiation physics and dosimetry that need to be addressed to determine the stopping power ratio. The term "stopping power ratio" is a crucial concept in dosimetry because it compares how different materials stop ionizing radiations, relatively, with respect to water which has been used as a standard reference because of its tissue equivalent properties. For accurate dose estimation as well as treatment planning within the radiation therapy field, it is important to consider this quotient. Using this formula calculates stopping powers for high-energy protons in both the material under consideration and the reference material on a theoretical basis only; these models incorporate such variables as mean excitation energy, atomic number, density etc. The PSTAR database and experimental measurements offer stopping power values for a variety of materials and energies. Theoretical computations are frequently validated by these empirical values. For protons in a variety of materials, including water, which is frequently used as the reference material, the PSTAR database from NIST offers stopping power and range tables. SPR is used in radiation therapy to translate dose measurements in a phantom substance (often water) into doses in real biological tissues or other materials. Precise SPR values guarantee accurate dosage administration to the intended tissues while preserving the health of nearby healthy tissues. To identify suitable dosimetry materials and create novel dosimetry techniques, SPR can be used to compare the radiological properties of various materials. To view the various energy sources' stopping power ratios by plotting the ratio via PSTAR water, we determined the SPR for each stopping power of different elements and compounds. An essential step in dosimetry that guarantees precise dose estimation and boosts radiation therapy efficacy is evaluating the stopping power ratio. SPR values for a variety of materials and applications can be achieved with accuracy and reliability by using modern tools such as the PSTAR database, theoretical models, and empirical data.

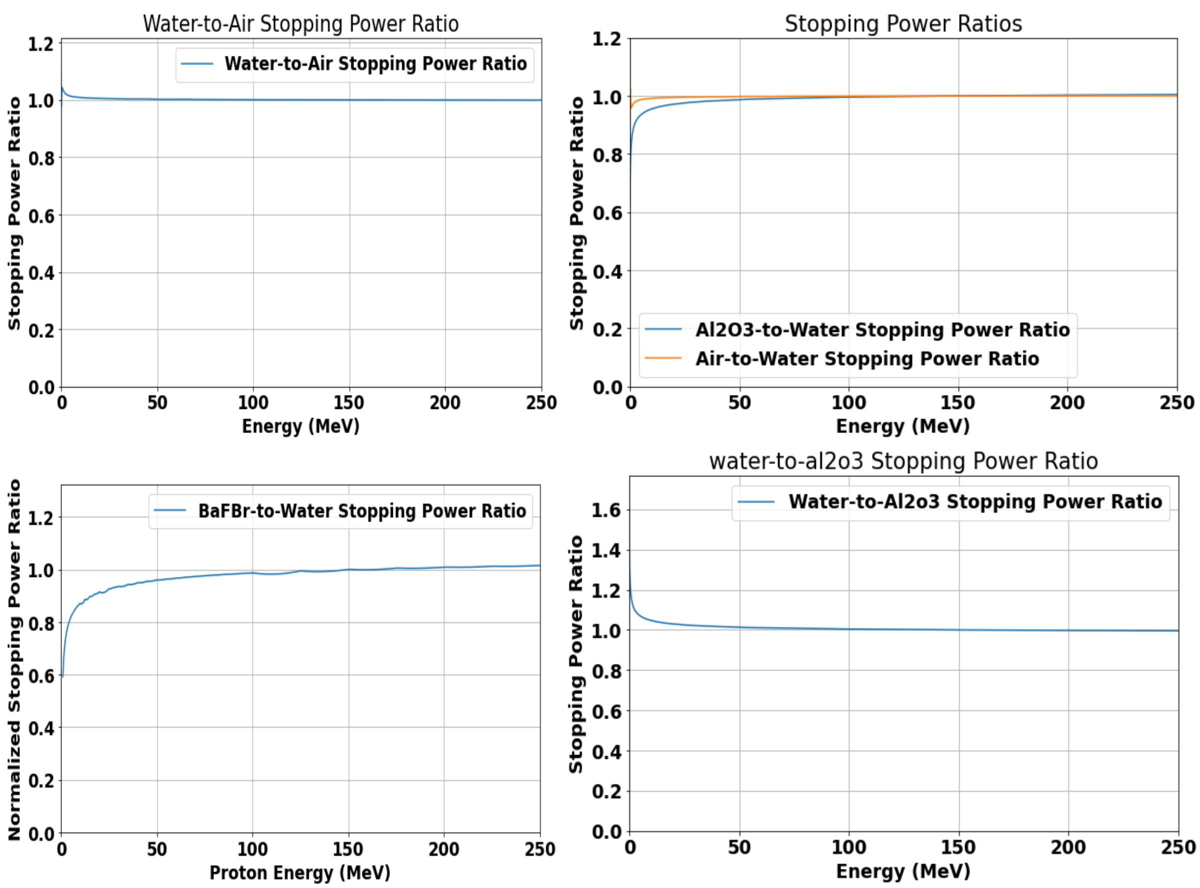

Figure 3: Graph illustrating the energy range for water/air, water/Al2O3, and BaFBr/water stopping power ratios. At 10 MeV, water had a stopping power ratio of about 1.05 to that of air. On the other hand, at 1 MeV it was roughly 1.12 in relation to Al2O3 while BaFBr to water ratio was approximately 0.95. These results are summarized in figures below by showing the calculated stopping power ratios for Al2O3-to-water and air-to-water and other pairs like BaFBr-to-water as well as water-to-Al2O3 based on our theoretical models. The findings also show that dosimeters are generally accurate in LET assessment at higher proton energies, therefore agreeing with predictions made by theory. However, there were large differences in lower energies where we used the semiempirical model as seen above. These observations underscore the limitation of current dosimeter techniques in measuring low-energy linear energy transfer (LET) as proposed by semiempirical modeling approach taken for this study. The PSTAR database provided stopper information for liquid water and air while Janni code was employed to determine data related to Al2O3 (PSTAR). To ensure comparability between different materials at different energies both water and air values were rescaled into those using protons of 150MeV energy scale (Janni, 1966) [18]. We normalized the stopping power data dividing them by corresponding values at each energy relative to the ones that they had at an incident proton beam with an energy of 150MeV (Eckerman et al., 2008)[19]. For direct comparison between stopping powers statistics for air, these statistics were interpolated to match percent depth dose value

Discussion

These results show promise for the use of high-Z_{eff} materials like BaFBr and low-Z_{eff} materials such as Al2O3 and water, in proton therapy for accurate LET measurements. Accurate dosimetric applications are possible since theoretical models that we developed concur with experimental data. These findings imply that were proton therapy dosimetry to be improved, there would be significant implications. This means that when designing dosimeters, it is important to consider the choice of material because slight variations in stopping power ratio across different materials indicate material selection impacts on dosimeter design. An example is from 10 MeV where a water-to-air ratio of 1.05 indicates small corrections will have to be made on air-filled ionization chambers used in measuring dose inside water-equivalent tissues

Conclusion

The article, affirming the accuracy in LET measurement by Bethe-Bloch theory and semiempirical models in proton therapy, is a major step towards validating the use of these models. When it comes to measuring LET levels with greater precision, dosimeters that are made from Al2O3, BaFBr and water may be employed. The work that must be done next will primarily focus on improving these models further as well as investigating their applicability in the clinical environment and examining different materials composition and states. This analysis gives a good understanding of stopping power prediction for material utilized in dosimetry and radiation physics applications today.

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